

System Instructions — Employee And Delivery Management System

Running the Program

The system automatically connects to the database upon startup, so no additional setup or data-loading arguments are required.

Ensure you have Java 21 or higher installed.

Running the Executable JAR

Open a terminal in the project root directory and run the following command:

```
java -jar adss2025_v01.jar
```

Using The System

Login

Upon launching, the system prompts for:

1. **Employee ID** — a 9-digit number
2. **Password** — at least 6 characters

The system automatically detects whether the user is an HR Manager a Transport manager or a regular Employee and routes them to the appropriate menu.

HR Global Dashboard

After logging in as an HR Manager, this top-level menu is displayed to manage branches and global settings. The following options are available:

- **Register New Branch** — Create a new branch in the system by entering its ID, name/location, physical address, phone number, contact person, and delivery zone details.
- **Select Branch to Manage** — Choose an existing branch to open its dedicated Branch

HR Menu.

- **Add Transport Manager** — Register a new Transport Manager to the system by entering their employee ID, name, and an initial password.
 - **Set Global Deadline** — Set the system-wide deadline date for employees to submit their weekly shift availability and preferences.
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HR Menu

Accessed through "Select Branch to Manage" from the Global Dashboard menu:

1. Shift Definition

- **Create Shift** — Select a day of the week and shift type (Morning/Evening) to create a new shift.
- **Set Requirements** — Select a shift, then specify how many employees of a given role (Cashier, Warehouse, Shift Manager) are required.

2. Shift Filling

- **View next week's schedule** — Displays all active shifts with their current assignments.
- **Assign employee** — Select a shift, role, and employee. If no available employees exist, the system offers to send an override request.
- **Unassign employee** — Remove an employee from a specific role in a shift.
- **View employee constraints** — Enter an employee ID to view their weekly availability submission.
- **View employee preferences** — Enter an employee ID to view their weekly preferences submission.

3. Employee Management

- **Hire employee** — Enter all employee details (ID, password, name, bank account, start date, employment type, salary type, salary, vacation days, overtime preference, day off, double shift preference, certifications).
- **Dismiss employee** — Enter employee ID to deactivate them.
- **Rehire employee** — Enter employee ID and new password to reactivate a dismissed employee.
- **View employee data** — Enter employee ID to view full profile.
- **Edit employee data** — Sub-menu with options:
 - Financial Details (bank account, salary type, salary)
 - Employment Details (employment type, vacation days, start date)
 - Name
 - Certifications (add or remove)

- Settings (day off, double shift preference, overtime preference)

4. Request Handling

- **View requests** — Displays all override requests sent by this HR manager. Select a request to view details. If the request has been approved by the employee, the HR manager can proceed to assign with override.

5. View History

- Enter a date and shift type to view details of a past (archived) shift.

6. View PENDING Deliveries Alerts

- Displays all PENDING deliveries created by the Transport Manager.
A delivery is marked PENDING if it is missing an assigned driver (with the required license) **or** a warehouse worker for the scheduled shift.

7. Back to Global Dashboard

- returns to the HR Global Dashboard screen.
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Employee Menu

After logging in as a regular employee, the following options are available:

1. Availability

- **View constraints** — Displays the employee's current weekly availability submission.
- **Set constraints** — For each day and shift of the upcoming week, the employee indicates whether they are available. Submitting an empty response clears all previous submissions.
- **View preferences** — Displays the employee's current weekly preferences.
- **Set preferences** — For each day and shift, the employee indicates preferred shifts.
Note: preferences can only be set for shifts where a constraint (availability) was already submitted.

2. Requests

- **View requests** — Displays all override requests received by this employee. The employee can select a request and choose to approve or reject it.

3. My Profile

- **View my details** — Displays the employee's full profile (ID, name, status, bank account, start date, employment type, salary, vacation days, day off, certifications).
- **Edit settings** — Allows the employee to update their day off, double shift preference, and overtime preference.

4. Logout

- Logs out the current user and returns to the login screen.
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Transport Menu

Accessed after logging in as a Transport Manager:

1. Deliveries

- **Create delivery** - Enter delivery details (ID, date, time), select source site, truck, and driver, then attach shipping documents. Status is set to READY if staff requirements are met, or PENDING otherwise.
- **Update delivery** - Edit details of an existing PLANNED, READY, or PENDING delivery.
- **View all deliveries** - Displays all registered deliveries.
- **View delivery details** - Select a delivery to view its routing profile and attached documents.
- **Execute delivery** - Select a READY or EXECUTING delivery to track its progress. At each checkpoint, record arrival weight or pause execution. If overweight, resolve via unloading, changing destination, or swapping truck.
- **Resolve PENDING deliveries** - Select a PENDING delivery to re-verify staff assignments.

2. Trucks

- **Add truck** - Register a new truck by entering license number, model, net weight, max capacity, and required license type (B, C, or C1).
- **View all trucks** - Displays all trucks in the fleet.
- **View truck details** - Select a truck to view its full technical specifications.

3. Drivers

- **View all drivers** - Displays all active drivers.
- **View driver details** - Select a driver to view their profile and license type.

4. Sites

- **Add Supplier Site** - Create a new external supplier site by entering ID, name, address, contact info, and delivery zone. (Note: company branches are managed via HR Global Dashboard).
- **View all sites** - Displays all registered branches and supplier sites.
- **View site details** - Select a site to view its full contact and zone details.

5. Back

- Return to the main menu.

Business Rules

- Each branch may have only one HR Manager, although the system supports multiple branches.
 - Every shift must have at least one Shift Manager.
 - An employee cannot be assigned to a shift on their designated day off.
 - An employee can only be assigned to a shift if they have submitted availability (a constraint) for that slot, unless an override request is approved.
 - A Shift Manager can hold up to two roles during the same shift.
 - A regular employee can be assigned to only one role per shift.
 - Preferences can only be submitted for shifts where availability has already been indicated.
 - Override requests must be approved by the employee before the HR Manager can assign them using an override.
 - If an employee does not submit their availability by the deadline, they are automatically considered available for all shifts that week (excluding their designated day off).
 - A driver's license must meet or exceed the truck's required license type. The hierarchy is:
 - B License: Can drive only B trucks.
 - C License: Can drive B and C trucks.
 - C1 License: Can drive B, C, and C1 trucks.
 - When a delivery is created, its initial status depends on availability: it is immediately set to **READY** if both a qualified driver and a warehouse worker are already assigned to the scheduled shift, or **PENDING** if either is missing. Once the truck departs and is actively tracked along its route, the status updates to **EXECUTING**. If the truck is weighed at any site during the trip and exceeds its maximum capacity, it is flagged as **OVERWEIGHT**. Finally, when the truck successfully finishes its entire route without issues, the delivery is marked as **COMPLETED**.
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Pre-loaded Data

Note on Data Initialization: Our system relies on time sensitive scheduling logic, where operations such as shift creation, constraint submissions, and transport planning are restricted to a rolling one week window. For example, the system prevents creating a shift for the 21st if the current date is only the 3rd. Because the validity of our scheduling data depends on the actual date the project is being tested, manual data entry was not feasible. Static data would quickly become outdated and trigger system errors. To ensure all dates remain relevant and functional, we implemented an automated initialization process that populates the database through code.

Users Table

ID	Password	IsLoggedIn
100000001	sarah123	0
100000002	yossi123	0
100000003	dana1234	0
100000004	ron12345	0
100000005	transport123	0

employee_id (Foreign Key): Links directly to the Employees table, associating this login credential with a specific worker.

password: Required for authentication.

IsLoggedIn: A status indicator that tracks whether the user is currently signed into the system.

System Logic & Data Connections

- The system determines the user's interface based on their unique ID and password: Sarah Cohen (ID 100000001, Password sarah123) is granted access to the HR Global Dashboard for personnel management services. Avi Transport (ID 100000005, Password transport123) is directed to the Transport Menu to manage logistics. All other users are routed to the Employee Menu, which provides personal access to submit constraints and preferences, view individual data, and perform other standard tasks.

Employee Table

id	name	bank_account	start_date	employment_type	salary_type	salary	vacation	will_overtime	status	branch_id
100000001	Sarah Cohen	100001	2025-01-15	FULL_TIME	GLOB AL	12000	20	1	ACTIVE	0
100000002	Yossi Levi	100002	2024-06-01	PART_TIME	HOURLY	45	10	0	ACTIVE	1
100000003	Dana Mizrahi	100003	2021-09-10	FULL_TIME	HOURLY	55	15	1	ACTIVE	1
100000004	Ron Shapiro	100004	2025-01-15	FULL_TIME	GLOB AL	10000	18	1	ACTIVE	1
100000005	Avi Transport	100005	2024-01-01	FULL_TIME	GLOB AL	12000	18	1	ACTIVE	0
100000006	David Driver	100006	2024-01-01	FULL_TIME	GLOB AL	11000	18	1	ACTIVE	1
100000007	Dana NoShift	100007	2024-01-01	FULL_TIME	GLOB AL	11000	18	1	ACTIVE	1

id (Primary Key): The unique identifier for every staff member.

name: The employee’s name.

bank_account: The employee’s bank account number.

start_date: The exact date the employee joined the store.

employment_type & salary_type: Defines the employee's contract (FULL_TIME, PART_TIME) and how they are paid (GLOBAL, HOURLY).

salary: The salary based on their salary_type.

vacation: The number of vacation days the employee currently has available.

will_overtime: A boolean flag (1 or 0) indicating whether the employee agrees to work overtime hours.

status: A flag (INACTIVE, ACTIVE) ensuring the algorithms only pull from currently employed workers.

branch_id (Foreign Key): Links directly to the Sites table, assigning the employee to its branch.

System Logic & Data Connections

- **Employees → Scheduling Algorithm:** The will_overtime column acts as a constraint. When an HR manager manually assigns an employee to a Morning (M) shift, the system prompts the manager to specify if the assignment includes overtime hours. If the manager selects "yes" but the employee's will_overtime status is 0, the system will block the assignment.
- **Employees → System Access:** The status column serves as a global security switch for the system. If an employee's status is INACTIVE, the system automatically denies them access to all services and prevents them from logging in entirely, until an HR manager manually updates their status back to ACTIVE.

Employee_Certifications Table

employee_id	certification
100000001	SHIFT_MANAGER
100000001	HR_MANAGER
100000002	WAREHOUSE
100000002	CASHIER
100000003	WAREHOUSE
100000003	CASHIER
100000004	SHIFT_MANAGER
100000004	CASHIER
100000005	DELIVERY_MANAGER
100000006	DRIVER
100000007	DRIVER

employee_id (Foreign Key): Links directly to the Employees table.

certification: Defines the professional roles an employee is authorized to perform (SHIFT_MANAGER, WAREHOUSE, CASHIER, DELIVERY_MANAGER, DRIVER, HR_MANAGER).

System Logic & Data Connections

Employee_Certifications → Scheduling: The scheduling algorithm checks this table before approving to assign a worker to a shift. For example, a user must have the DELIVERY_MANAGER certification to manage transport routes, or the DRIVER certification to be assigned to a vehicle in the Deliveries table.

Slot_Submission Table (Sample)

Because each employee submits 14 distinct records per week (2 shifts per day over 7 days), this table grows rapidly. To maintain document readability, only a representative sample of submissions for two employees is shown below.

employee_id	date	shift_type	constraint_val	preference
100000002	2026-07-09	MORNING	1	0
100000002	2026-07-09	EVENING	0	0
100000002	2026-07-08	MORNING	0	0
100000002	2026-07-08	EVENING	0	0
100000003	2026-07-11	MORNING	1	1
100000003	2026-07-11	EVENING	1	1
100000003	2026-07-10	MORNING	0	0

employee_id (Foreign Key): Links to the Employees table.

date: The specific calendar date for the shift.

shift_type: Defines the time block (MORNING, EVENING).

constraint_val: A binary indicator (1 or 0) representing if an employee is available for that specific shift.

preference: A binary indicator (1 or 0) representing whether the employee prefers to work that specific shift.

System Logic & Data Connections

- **Slot_Submission → Scheduling Algorithm:** The scheduling algorithm uses this table to map the workforce's weekly availability. For example, the data shows that employee 100000002 (Yossi Levi) has submitted availability for the MORNING shift on July 9th, but his preference value of 0 indicates he does not favor that shift and would prefer to be assigned elsewhere.
- **Slot_Submission → Shifts:** This table acts as a filter for scheduling. If an employee submits a 0 for constraint_val, the system is strictly instructed not to authorise assigning that worker to that specific date and shift type.

Weekly_Submission Table

employee_id	day_off	double_shift_allowed
100000001	1	0
100000002	7	0
100000003	6	1
100000004	2	1
100000005	1	0
100000006	1	0
100000007	1	0

employee_id (Foreign Key): Links directly to the Employees table.

day_off: Specifies the designated day of the week (1–7) on which the employee is not available for work.

double_shift_allowed: A binary flag indicating if the employee agrees to work two shifts consecutively.

System Logic & Data Connections

Weekly_Submission → Override_Requests: If an HR manager wants to assign an employee to a double shift but the employee's double_shift_allowed flag is 0 or it's on the employee's

day_off, the system prevents the assignment. In such cases, the HR manager must submit an override request to bypass the restriction.

Shifts Table

id	branch_id	date	shift_type	is_active
SHIFT_001	1	2026-07-07	MORNING	1
SHIFT_001_02	2	2026-07-07	MORNING	1
SHIFT_001_04	4	2026-07-07	MORNING	1
SHIFT_002	1	2026-07-08	MORNING	1
SHIFT_003	1	2026-07-09	MORNING	1
SHIFT_004	1	2026-01-15	MORNING	0

id (Primary Key): A unique identifier for the shift.

branch_id (Foreign Key): Links the shift to a specific branch in the network.

date & shift_type: Defines the time of the shift.

is_active: A boolean flag (1 or 0) indicating if the shift is currently in the active scheduling rotation or if it has been archived.

System Logic & Data Connections

- **Shifts → Sites:** The branch_id handles distributed scheduling. For example, on July 7th, multiple morning shifts (SHIFT_001, 02, 04) run simultaneously across the Beer Sheva, Dimona, and Eilat branches.
- **Shifts → Shift_Required_Roles:** The shift ID acts as a key to its specific staffing matrix. SHIFT_001 requires exactly 1 Warehouse, 1 Manager, 1 Driver, and 2 Cashiers.

- **Shifts → Shift_Assignments:** We can see that for example, in SHIFT_001, Ron Shapiro is assigned to both Shift Manager and Cashier.
- **Shifts → Shift Archive Display:** The system maintains an archive of historical shifts, allowing managers to review past shifts. Here we can see a specific shift SHIFT_004, where the is_active value is set 0 (acts as an archive indicator).

Shift_Required_Roles Table

shift_id	certification	required_count
SHIFT_001	WAREHOUSE	1
SHIFT_001	SHIFT_MANAGER	1
SHIFT_001	DRIVER	1
SHIFT_001	CASHIER	2
SHIFT_001_02	WAREHOUSE	1
SHIFT_001_02	SHIFT_MANAGER	1
SHIFT_001_02	DRIVER	1
SHIFT_001_02	CASHIER	2
SHIFT_001_04	WAREHOUSE	1
SHIFT_001_04	SHIFT_MANAGER	1
SHIFT_001_04	DRIVER	1
SHIFT_001_04	CASHIER	1

SHIFT_002	SHIFT_MANAGER	1
SHIFT_002	CASHIER	2
SHIFT_003	WAREHOUSE	1
SHIFT_003	SHIFT_MANAGER	1
SHIFT_003	CASHIER	1
SHIFT_004	WAREHOUSE	1
SHIFT_004	SHIFT_MANAGER	1
SHIFT_004	CASHIER	1

shift_id (Foreign Key): Links directly to a specific shift in the Shifts table, connecting the requirement to a specific date and branch.

certification: Specifies the exact job role required.

required_count: An integer defining exactly how many employees are needed to that specific certification.

Shift_Assignments Table

shift_id	certification	employee_id
SHIFT_001	WAREHOUSE	100000003
SHIFT_001	SHIFT_MANAGER	100000004
SHIFT_001	CASHIER	100000004
SHIFT_001_02	WAREHOUSE	100000002
SHIFT_001_02	DRIVER	100000006
SHIFT_001_04	WAREHOUSE	100000002
SHIFT_001_04	DRIVER	100000006
SHIFT_002	SHIFT_MANAGER	100000001
SHIFT_002	CASHIER	100000003
SHIFT_003	WAREHOUSE	100000003
SHIFT_003	SHIFT_MANAGER	100000004
SHIFT_003	CASHIER	100000002

SHIFT_004	WAREHOUSE	100000003
SHIFT_004	SHIFT_MANAGER	100000004
SHIFT_004	CASHIER	100000002

shift_id (Foreign Key): Links directly to a specific shift in the Shifts table, linking the assignment to a specific date and branch.

certification: Defines the specific job role that is fulfilled during that shift.

employee_id (Foreign Key): Links directly to the Employees table, identifying the employee who has been scheduled to fulfill that required role.

Override_Requests Table

id	hr_id	emp_id	date	shift_type	role	status
REQ_1	100000001	100000002	2026-07-08	M	CASHIER	PENDING

id (Primary Key): A unique identifier for the specific override request.

hr_id (Foreign Key): Links to the Employees table, identifying the manager who initiated the override.

emp_id (Foreign Key): Links to the Employees table, identifying the worker being requested outside of their normal availability constraints.

date & shift_type: Specifies the exact time container the manager wants to assign the employee to.

role: Defines the specific job role the employee is being asked to fulfill.

status: Tracks the current state of the request in the workflow (PENDING, APPROVED, REJECTED).

System Logic & Data Connections

- **Override_Requests → Employees:** The hr_id and emp_id link directly to the Employees table. For example, the data shows that manager Sarah Cohen initiated an override request for Yossi Levi to work as a Cashier during the morning shift on 2026-07-08.

Deliveries Table

ID	Date	Departure Time	Recorded Weight	Status	Source SiteId	TruckLicense Number	DriverId	Current Step	PendingReason
1	2026-07-07	08:30	4000.0	READY	1	123-45-678	100000006	0	NULL
2	2026-07-07	09:00	500.0	PENDING	1	222-33-444	100000007	0	Delivery rejected: Driver ID ...

ID (Primary Key): A unique identifier for the specific delivery.

Date & DepartureTime: The exact scheduled launch window for the route.

RecordedWeight: The calculated total weight of all cargo loaded for the trip, necessary for vehicle capacity validation.

Status, CurrentStep, & PendingReason: these help tracking real time progress. The Status manages the delivery lifecycle: it initializes as **READY** (if both a qualified driver and warehouse worker are already assigned) or **PENDING** (if staffing is missing). Once dispatched, it updates to **EXECUTING**. If a mid-route weigh-in exceeds truck capacity, it flags as **OVERWEIGHT**, and upon finishing the route, it is marked as **COMPLETED**. The CurrentStep acts as a saved index for the route's destination sites, because the transport manager executes deliveries sequentially (arriving, weighing, and choosing to stop or continue), this column is vital for resuming paused deliveries exactly where they left off. PendingReason captures specific errors if a dispatch is blocked.

SourceSiteId (Foreign Key): Links to the Sites table, establishing the physical starting point of the route.

TruckLicenseNumber & DriverId (Foreign Keys): Links to the Trucks and Employees tables, locking in the specific truck and driver assigned to the delivery.

System Logic & Data Connections

- **Deliveries → Trucks & Employees:** For example, the data shows Delivery 1 successfully pairing Driver 100000006 (David Driver) with Truck 123-45-678 (a Volvo FL Series).

- **Deliveries** → **DeliveryDocuments** → **Sites**: For example, Delivery 1 originates at SourceSiteId 1 (Beer Sheva Store Branch). By tracing this delivery through the DeliveryDocuments table, we can see the route with scheduled drop-offs at Sites 2 (Dimona), 3 (Tnuva Dairy Supplier), and 4 (Eilat).

DeliveryDocuments Table

DocumentId	DeliveryId	DestinationSiteId
1001	1	2
1002	1	3
1003	1	4
1004	2	2

DocumentId (Primary Key): A unique identifier for the specific delivery document, representing a distinct order or drop-off point.

DeliveryId (Foreign Key): Links back to the Deliveries table. It lets the system group several different store drop-offs together to one single delivery route.

DestinationSiteId (Foreign Key): Links directly to the Sites table, specifying the exact physical location (such as the Dimona or Eilat branch) of the site associated with this delivery document..

Drivers Table

ID	LicenseType
1000000006	B
1000000007	B

ID (Primary/Foreign Key): Links directly to the Employees table, acting as the unique identifier for the worker.

LicenseType: Specifies the exact legal driving classification the employee currently holds (B, C1, C).

System Logic & Data Connections

- **Drivers → Employees:** This table acts as an extension of the main HR database. The system uses the ID to attach transport-specific data only to relevant staff. For example, employee 1000000006 (David) exists in the main system, but this table specifically marks him as a certified driver.

Sites Table

ID	Name	Address	PhoneNumber	ContactPerson	SiteType	Zoneld	ZoneName
1	Beer Sheva Store Branch	Bar Nisan 6	0587243922	Alex Roso	BRANCH	1	South Zone 101
2	Dimona Store Branch	Rager 6	0587243922	Bar Bussani	BRANCH	1	South Zone 102
3	Tnuva Dairy Supplier	Avraham Avinu 10	0587243922	Omer Biton	SUPPLIER	1	South Zone 103
4	Eilat Store Branch	HaTmarim Blvd 40	0587243922	Noam Levi	BRANCH	2	South Zone 104

ID (Primary Key): A unique identifier for the specific physical location.

Name & Address: The designated title and physical street location of the branch or supplier site.

PhoneNumber & ContactPerson: The direct contact details and the name of the person responsible for logistics.

SiteType: Explicitly distinguishing whether the site is an internal BRANCH or an external SUPPLIER.

Zoneld & ZoneName: Defines the broader geographic region the site belongs to.

Trucks Table

LicenseNumber	Model	NetWeight	MaxCapacityWeight	RequiredLicenseType
123-45-678	Volvo FL Series	3000.0	8000.0	B
222-33-444	Ford Transit Heavy Utility	3200.0	10000.0	B
987-65-432	Isuzu Forward	4500.0	12000.0	C1
555-12-345	Scania R-Series	7500.0	26000.0	C

LicenseNumber (Primary Key): The unique identifier for the vehicle, which is its physical license plate number.

Model: The model of the vehicle.

NetWeight: The base, empty weight of the truck itself.

MaxCapacityWeight: The maximum payload limit the truck can safely transport, which is important for the routing algorithm's weight calculations.

RequiredLicenseType: The exact legal driving classification (B, C1, C) that a driver must hold in order to be authorized by the system to operate this specific vehicle.

TransportedItems Table

ItemId	DocumentId	ItemName	Quantity
501	1001	Milk 3%	100
502	1001	White Bread	50
503	1002	Yellow Cheese	30
503	1003	Yellow Cheese	30
504	1004	Yogurt	20

ItemId & ItemName: The unique identifier and description of the specific product being shipped.

DocumentId (Foreign Key): Links directly to the DeliveryDocuments table, attaching these specific products to a distinct delivery document.

Quantity: The exact number of units of the item that need to be delivered for that specific document.

System Logic & Data Connections

- **TransportedItems → DeliveryDocuments → Sites:** This connection tracks exactly what inventory is going where. For example, the data shows that Document 1001 contains exactly 100 units of Milk (Item 501) and 50 units of White Bread (Item 502). By tracing Document 1001 back to the DeliveryDocuments table, we know it is destined for Site 2.